MIL-B-43481E
31 May 1979
SUPERSEDING
MIL-B-43481D
10 March 1977

MILITARY SPECIFICATION

BOOTS, COMBAT, LEATHER, BLACK, DIRECT MOLDED SOLE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for a full-laced mildew resistant leather combat boot with a direct molded sole.

1.2 Classification. The boots shall be of one type in the following sizes and widths as specified (see 6.2).

TABLE I. Schedule of sizes

* Width	3	3½	4	4½	5	5½	6	6½	7	7½	8	8½	9	9½	10	10½	11	11½	12	12½	13	13½	14
XN	-	-	-	-	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
N	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
R	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
W	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
XW	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Natick Research and Development Command, Natick, MA 01760 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8430

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2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

V-L-61	-	Laces, Nylon.
V-T-295	-	Thread, Nylon.
KK-I-570	-	Insole, Footwear, Leather, Cattlehide.
TT-C-490	-	Cleaning Methods and Pretreatment of Ferrous Surfaces for Organic Coatings.
UU-P-268	-	Paper, Kraft, Wrapping.
DDD-T-86	-	Tape, Textile; Cotton, General Purpose (Unbleached, Bleached, or Dyed).
GGG-N-350	-	Nippers and Pincers.
PPP-B-636	-	Boxes, Shipping, Fiberboard.

MILITARY

MIL-I-1821	-	Insoles, Footwear (Ventilating Type).
MIL-L-3122	-	Leather, Cattlehide, for Footwear Uppers, Chrome Tanned, Fatliquored.
MIL-L-10867	-	Leather, Cattlehide, Gusset, Chrome Tanned, Fatliquored.
MIL-C-13924	-	Coating, Oxide, Black, for Ferrous Metals.
MIL-C-41814	-	Counter, Footwear.
MIL-L-41815	-	Lasts, Footwear, Men's, Combat, U.S., MIL-5.
MIL-C-43956	-	Cloth, Twill, Cotton, 10.0 Ounces (339 g).

STANDARDS

FEDERAL

FED-STD-311	-	Leather, Methods of Sampling and Testing.
FED-STD-601	-	Rubber: Sampling and Testing.
FED-STD-751	-	Stitches, Seams, and Stitchings.

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MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage.

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DRAWING

US ARMY NATICK RESEARCH AND DEVELOPMENT COMMAND

2-1-884 - Chevron Design Sole for DMS Leather Combat Boot.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with the specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

B 36	- Brass Plate, Sheet, Strip, and Rolled Bar.
D 518	- Surface Cracking Resistance of Stretched Rubber Compound.
D 573	- Accelerated Aging of Vulcanized Rubber by the Oven Method.
D 1052	- Measuring Cut Growth by the Use of the Ross Flexing Machine.
D 1149	- Accelerated Ozone Cracking of Vulcanized Rubber.
D 1630	- Abrasion Resistance of Rubber Compounds for Soles and Heels.
D 2240	- Indentation Hardness of Rubber and Plastics by Means of a Durometer.
E 18	- Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials.

(Application for copies should be addressed to the American Society for Testing Materials, 1916 Race Street, Philadelphia, PA 19103.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, N.W., Washington, DC 20036.)

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UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Guide sample. Samples, when furnished, are solely for guidance and information to the contractor (see 6.3). Variation from the specification may appear in the sample, in which case the specification shall govern.

3.2 Laboratory report and first article approval.

3.2.1 Laboratory report approval. Unless otherwise specified (see 6.2), the bidder shall submit to the contracting officer for approval, a certified copy of a recent laboratory report on the vulcanized footwear which he intends to deliver. The report shall certify that the rubber compound and the vulcanizing adhesive which he proposes to use provides the required strength of bond and conforms to the other requirements of this specification. The purpose of the above requirement is to assist the Government in determining the capability of contractors to manufacture a boot meeting all the requirements of the specification. The submission of an acceptable report under this requirement shall not be construed as relieving a contractor from subsequently meeting all requirements of the specification on all deliveries.

3.2.2 First article. When specified (see 6.2), the contractor shall furnish a sample for first article inspection and approval (see 4.3 and 6.5).

* 3.3 Material (see 6.6).

3.3.1 Leather. All leather components covered by this specification shall contain not less than 0.18 percent nor more than 0.70 percent paranitrophenol fungicide.

3.3.1.1 Upper leather vamps, quarters, and one piece counterpocket-backstays. The leather for the vamps, quarters, and one piece counterpocket-backstays shall be smooth plated, full grain or corrected grain side leather conforming to treatment A of MIL-L-3122 except that the thickness shall be as specified for the cut parts in 3.7.1 and the counterpocket-backstay may be split or shaved

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to the specified thickness. The break for vamps and quarters shall not be more than a break pattern of 4 when examined as specified in 4.4.3.1.1. The color shall be black. All finishes applied by the boot manufacturer and tanner shall be compatible with each other to insure a long-lasting lustrous finish.

3.3.1.2 Leather tongues, eyelet stays, and top facings. Tongues, eyelet stays and top facings shall be gusset leather conforming to treatment A of MIL-L-10867, except that the thickness shall be as specified in 3.7.1 and that thinner pliable portions of leather conforming to MIL-L-3122 and split to a thickness of not less than 2-1/4 ounces (0.9 mm) nor more than 3-1/4 ounces (1.3 mm) may be used for eyelet stays and top facings. The color shall be black.

3.3.1.3 Leather insoles. The leather insoles shall conform to type I, tannage a or b, class 1 of KK-I-570. The insoles shall be 7 irons (4 mm) in thickness. The insoles for this boot contain no channel area.

3.3.1.4 Counters. Counters shall conform to the requirements of MIL-C-41814 for boots, combat, leather, black, DMS.

3.3.2 Fabrics.

3.3.2.1 Backseam tape. The tape shall conform to type I, class 1, 1/2 or 17/32 inch (13 or 14 mm) width of DDD-T-86.

3.3.2.2 Box toes. The box toes shall be cut from a material consisting of a woven base flannel or a processed (needled) fabric impregnated with a pyroxylin or polystyrene compound. The box toes may be either an activated or a prepared type. Thickness of this material shall be 0.0875 ± 0.0075 inch (2.22 ± 0.19 mm) when tested as specified in 4.4.1. The box toes shall be of a suitable shape and size to engage the insoles around their toe radius after lasting, and shall be skived 9/16 ± 1/8 inch (14 ± 3 mm) wide at the breast.

3.3.2.3 Vamp lining. The vamp lining shall be cut from cotton twill conforming to class 2 of MIL-C-43956.

3.3.3 Metal.

* 3.3.3.1 Steel shank. The steel shank shall be constructed from 19 gage, 0.0418 ± 0.0030 inch (1.062 ± 0.076 mm) cold rolled carbon steel with a hardness ranging from 47 to 54 Rockwell C scale and shaped to conform to the arch of the MIL-5 last. The width of the shank for boot sizes 3 through 4 shall be 5/8 ± 1/32 inch (16 ± 1 mm) and for boot sizes 4-1/2 through 14 shall be 1 ± 1/32 inch

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(25 ± 1 mm). The shank shall be made with two ribs for the 5/8 inch (16 mm) width and three ribs for the 1 inch (25 mm) width and the overall height of each rib shall be 0.080 to 0.125 inch (2.03 to 3.18 mm). Each rib shall taper off to end within 1 ± 1/8 inch (25 ± 3 mm) from the toe end and within 1 ± 1/8 inch (25 ± 3 mm) from the heel end. The 1 inch (25 mm) shank shall have two prongs and the 5/8 inch (16 mm) shank shall have one prong 3/16 inch (5 mm) long inward into the filler and both shank widths shall have two prongs 3/32 inch to 5/32 inch (2 to 4 mm) long downward into the innersole. The steel shank shall have a zinc electroplated coating with dull or bright finish, or a zinc coating, completely and uniformly applied to the base metal following a thorough cleansing by any method of TT-C-490. Steel shanks with black oxide coating conforming to class 1 of MIL-C-13924 may be used in lieu of shanks with zinc coating. The shank length grading shall be as specified in table II. Testing shall be as specified in 4.4.1.

TABLE II. Shank lengths

All widths boot sizes	3 - 6½	7 - 8½	9 -10½	11 - 12½	13 - 14
Shank length, inches (mm ± 1/32 inch (± 1 mm) tolerance	4-1/8 (105)	4-3/8 (111)	4-5/8 (117)	4-7/8 (124)	5-1/8 (130)

3.3.3.2 Tacks and staple wire. The heel seat lasting tacks shall be brass or steel. The brass tacks shall be cut from alloy No. 6 or No. 8 of ASTM B 36. The side lasting staples and toe lasting staples shall be steel. Tacks shall be of sufficient length to attach the parts through which they are driven and clinch on the insole.

3.3.3.3 Eyelets. The eyelets shall be made of aluminum, 0.0160 ± 0.0015 inch (0.406 ± 0.038 mm) thick, with roll setting barrel and shall conform to the following finished requirements when tested as specified in 4.4.1.

Outside diameter of flange - 0.480 to 0.496 inch (12.19 to 12.60 mm)
Outside diameter of barrel - 0.295 to 0.299 inch (7.49 to 7.59 mm)
Diameter of hole before setting - 0.235 to 0.240 inch (5.97 to 6.10 mm)
Overall length - 0.224 to 0.236 inch (5.69 to 5.99 mm)

3.3.3.3.1 Finish of eyelets. After fabrication, the eyelets shall be anodized and top roller coated or tumble coated with not less than two coats of black enamel.

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* 3.3.4 Rubber compounds.

3.3.4.1 Rubber compound for outersole. The rubber compound for outersole shall conform to following requirements when tested as specified in 4.4.1 and 4.4.3.2: (a) through (h).

(a) Color. The color shall be black nonmarking producing only a slight erasable mark on white paper.
(b) Abrasive index. The abrasive index shall be not less than 175 before and after aging for 70 hours at 100 ± 1°C.
(c) Hardness. The durometer hardness shall be 65 ± 5. After accelerated aging for 70 hours at 100 ± 1°C, the durometer hardness shall change no more than 10 points from the original.
(d) Hardness and hardness change at -18°C. The hardness after one hour exposure in a cold chamber at -18°C shall be 80 maximum, and shall not exceed the hardness at room temperature by more than 20 points.
(e) Cut growth. The cut growth after accelerated aging for 70 hours at 100 ± 1°C, shall be not greater than 200 percent after 50,000 cycles.
(f) Ozone resistance. There shall be no sign of cracking after test for ozone resistance.
(g) Volume swell. The volume swell shall be not greater than 60 percent in medium No. 6 fuel.
(h) Bond strength. The bond strength shall be not less than 135 pounds (600 N) except that one specimen out of five tested may be below 135 pounds (600 N) but shall be not less than 120 pounds (530 N). If any specimens in the sample show clear evidence of tearing into the leather, those specimens shall be rejected and new ones selected for the test.

* 3.3.4.2 Rubber compound for heel. The rubber compound for heel shall conform to following requirements when tested as specified in 4.4.1 and 4.4.3.2: (a) through (f).

(a) Color. The color shall be black non-marking producing only a slight erasable mark on white paper.
(b) Abrasive index. The abrasive index shall be not less than 300 before and after aging for 70 hours at 100 ± 1°C.
(c) Hardness. The durometer hardness shall be 70 ± 5. After accelerated aging for 70 hours at 100 ± 1°C, the durometer hardness shall change no more than 10 points from the original.
(d) Hardness and hardness change at -18°C. The durometer hardness after one hour exposure in a cold chamber at -18°C shall be 90 maximum, and shall not exceed the hardness at room temperature by more than 20 points.
(e) Ozone resistance. There shall be no sign of cracking after test for ozone resistance.
(f) Volume swell. The volume swell shall be not greater than 60 percent in medium No. 6 fuel.

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3.3.4.3 Bottom design. The bottom design shall be for a chevron type sole and heel tread conforming to Drawing 2-1-884. Sole thickness shall be as specified on the drawing and shall match within 1/16 inch (2 mm) between left and right boots.

3.3.4.4 Polyvinyl chloride exclusion. The rubber for the outsole and heel shall contain no polyvinyl chloride (PVC).

3.3.5 Thread.

* 3.3.5.1 Thread, upper fitting. The thread for upper fitting shall be nylon and shall conform to types I, II, or III of V-T-295 except that the requirements for colorfastness shall not apply. The thread sizes shall be E for needle and bobbin with stitch type 301, and E for the needle and B or E for the looper with stitch type 401. The color of the thread shall be black.

3.3.6 Miscellaneous materials.

3.3.6.1 Heel fillers. The heel fillers shall be made of fiberboard or other suitable material that does not affect vulcanization of the heel. The fillers shall be heel-shaped with straight edges. The heel fillers shall conform to the dimensions of table III when tested as specified in 4.4.1.

TABLE III. Dimensions of heel fillers, inches (mm)

Heel filler size designation	A	B	C	D
Heel curve radius 1/	7/16 (11)	9/16 (14)	11/16 (17)	13/16 (21)
Width 2/	7/8 (22)	1-1/8 (29)	1-3/8 (35)	1-5/8 (41)
Length 1/	2 (51)	2-1/8 (54)	2-1/4 (57)	2-3/8 (60)
Thickness 3/	17/32 to 25/32 (13 to 20)	17/32 to 25/32 (13 to 20)	17/32 to 25/32 (13 to 20)	17/32 to 25/32 (13 to 20)

1/ Tolerance of ± 1/32 inch (± 1 mm) is permitted.

2/ Tolerance of + 1/8 inch, - 1/32 inch (+ 3 mm, - 1 mm) is permitted.

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3/ There shall be a minimum of 5/16 inch (8 mm) rubber tread underneath the filler.

Heel fillers are to be used with boot sizes and widths as follows:

* Boot width	Boot sizes																									
	3	3½	4	4½	5	5½	6	6½	7	7½	8	8½	9	9½	10	10½	11	11½	12	12½	13	13½	14
XN	-	-	-	-	A	A	A	A	B	B	B	B	B	B	C	C	C	C	C	C	D	D	D
N	A	A	A	A	A	A	A	A	B	B	B	B	B	B	C	C	C	C	C	C	D	D	D
R	A	A	A	A	A	A	A	A	B	B	B	B	B	B	C	C	C	C	C	C	D	D	D
W	A	A	A	A	A	A	A	A	B	B	B	B	B	B	C	C	C	C	C	C	D	D	D
XW	A	A	A	A	A	A	A	A	B	B	B	B	B	B	C	C	C	C	C	C	D	D	D

3.3.6.2 Adhesives. The adhesives used for bonding the various leather and fabric components of the boots shall be of the following types:

(a) Natural rubber latex
(b) Synthetic rubber latex (including chloroprene)
(c) Natural rubber solvent cement
(d) Synthetic rubber solvent cement
(e) Synthetic resin cement

3.3.6.3 Laces. The laces shall conform to type II, class 3 of V-L-61. The color of the laces shall be black. The length of each lace shall be a minimum of 60 inches (152 cm).

3.3.6.4 Insole, ventilating. Each pair of boots shall be supplied with one pair of ventilating insoles conforming to MIL-I-1821. The ventilating insole size relation to applicable size boot shall be as indicated in MIL-I-1821. The following is an example of an insole size relationship to the applicable size boot as indicated in MIL-I-1821.

Sizes (boots)	Size (insoles)		
	Boot widths XN, N	Boot width R	Boot widths W, XW
8, 8½	8 N	8 R	8 W

3.3.6.5 Finishing materials for the leather uppers.

3.3.6.5.1 Repairers. Repairers shall be liquid spray, or crayon or paste type applied by hand. The color shall match the upper leather and shall have sufficient coverage to correct minor surface imperfections of the leather.

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3.3.6.5.2 Fillers. Fillers, where used, shall be capable of providing a suitable foundation for the application of the top finish and shall be applied by sponge or spray method.

3.3.6.5.3 Top finish. The top finish shall be applied by either the sponge or spray method, and shall be a material compatible with and capable of wetting into and binding the previously applied coats. The finish shall match the dyed leather and shall provide luster.

3.4 Design. The design shall be for a full lace closure with "V" notch in the eyelet row edge of the quarters, direct molded sole construction boot with removable ventilating insole, grain out leather upper, plain toe vamp, cloth lined vamp and box toe, one piece leather counterpocket-backstay, and full transverse chevron outersole and heel.

* 3.5 Patterns and dies. The height of the finished boot, measured upward on outside from tread area at breast of heel to top of boot, shall be 10-1/2 ± 1/4 inches (267 ± 6 mm) on size 9 and shall graduate up or down between whole sizes approximately 1/8 inch (3 mm) as indicated by patterns. A standard set of paper patterns will be furnished by the Government as a basis from which the contractor's dies and patterns shall be made. The patterns, and contractor's cutting dies and patterns shall consist of the component parts specified in table IV. The Government patterns shall be followed except for lasting allowance, which shall be determined by the contractor to provide a minimum of 1/2 inch (13 mm) lasting allowance. Patterns and markers may be slightly modified to aid in fitting during manufacture. The lastinng allowance around the heel seat may include two "V" notches no greater than 7/16 inch (11 mm) in depth at the center back area of the counter pocket. The contractor shall furnish the wood insole rounding patterns conforming to paper patterns loaned by the procuring activity. The one piece counterpocket-backstay may be used with an inverted "V" removed in the center of the back of the counterpocket. The height of the cut for this inverted "V" shall be a maximum of 1-1/4 inches (32 mm) from the bottom of the lasting allowance on a size 3 and grade upward to a maximum height of 1-1/2 inches (38 mm) on a size 14. The point of this inverted "V" shall be in the center back of the counter-pocket. All outer lines of this one piece counterpocket-backstay must conform to Government furnished patterns except for this inverted "V". This "V" shall be removed in a curved line to conform, after closing, to the back part of the MIL-5 last.

TABLE IV. Contractor's dies and patterns

Component parts	
Vamp	Whole and half sizes, all widths
Quarter ("V" notch)	Whole and half sizes, all widths
Counterpocket-backstay (one piece)	Whole and half sizes, all widths
Tongue	Whole and half sizes, all widths

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TABLE IV. Contractor's dies and patterns

Component parts	
Top facing	Five sizes, one width
Eyelet stay	Three sizes, one width
Vamp lining (cloth)	Whole sizes, all widths
Insole	Whole and half sizes, all widths
Box toe	Eight sizes, all widths

3.6.1 Lasts and markers. The boots shall be lasted on the MIL-5 last. All necessary sizes will be loaned to the contractor by the Government. A marker set consisting of a one piece counterpocket-backstay, a vamp, and a quarter, in whole and half sizes, all widths, will be loaned to the contractor by the Government.

3.6.2 Molds. The molds shall be made so as to produce an outersole conforming to Drawing 2-1-884 when tested as specified in 4.4.2.1.

3.7 Construction.

3.7.1 Cutting uppers. The uppers shall be cut from grain out leather. The leather shall conform to the thickness specified in table V. Except for eyelet stay, top facing, and one piece counterpocket-backstay, none of the upper parts shall be split or shaved. No parts cut off stretch shall be accepted. Vamps shall be cut from the bend area of the side leather.

TABLE V. Thickness of upper leather

Part	Thickness, ounces		Thickness, millimeters	
	Minimum	Maximum	Minimum	Maximum
Vamp	5	6	2.0	2.4
Quarter	4-1/2	6	1.8	2.4
Counterpocket-backstay (one piece)	4	6	1.6	2.4
Eyelet stay and top facing	2-1/4	3-1/4	0.9	1.3
Tongue	2-3/4	3-1/2	1.1	1.4

3.7.1.1 Skiving. The skiving of upper leather parts shall conform to the requirements specified in table VI.

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TABLE VI. Skiving requirements

Part	Side	Location	Scarf width, inches (mm) 6 ounce (2.4 mm) leather	5 ounce (2.0 mm) leather	4-1/2 ounce (1.8 mm) leather	4 ounce (1.6 mm) leather	Scarf edge thickness, millimeters ± 0.3 mm
Quarter	Flesh	Back edge	3/16 ± 1/32 (5 ± 1)	-	1/16 ± 1/64 (1.6 ± 0.4)	-	1.5
Quarter	Flesh	Lasting edge 1/	5/8 ± 1/8 (16 ± 3)	-	13/64 ± 3/64 (5.2 ± 1.2)	-	1.5
Quarter	Flesh	Top edge and eyelet row edge	3/16 ± 1/32 (5 ± 1)	-	3/32 ± 1/64 (1.6 ± 0.4)	-	1.3
Quarter	Flesh	Vamp margin 1/	3/16 ± 1/32 (5 ± 1)	-	1/16 ± 1/64 (1.6 ± 0.4)	-	1.5
Vamp	Flesh	Throat and wings	3/16 ± 1/32 (5 ± 1)	1/8 ± 1/32 (3 ± 1)	-	-	1.2
Vamp	Flesh	Toe 1/	3/4 ± 1/8 (19 ± 3)	9/16 ± 3/32 (14 ± 2)	-	-	1.5
Counter-pocket	Flesh	Lasting edge 1/ 2/	5/8 ± 1/8 (16 ± 3)	-	-	3/16 ± 1/64 (5 ± 0.4)	1.5
Counter-pocket-backstay	Flesh	Stitch edge 2/	3/16 ± 1/32 (5 ± 1)	-	-	1/16 ± 1/32 (1.6 ± 1)	1.3

1/ Skiving of the lasting edge of the counterpocket, toe vamp, quarter and quarter vamp margin is optional.

2/ When one piece counterpocket-backstay is 4 to 5 ounces (1.6 to 2.0 mm) thick, the skiving of the lasting counterpocket edge and stitch edge of the one piece counterpocket-backstay shall not apply.

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NOTE: A dash indicates that the scarf width is the same as indicated in the preceding column.

3.7.2 Vamp crimping. The vamps may be crimped, if necessary, to insure tight lasting.

* 3.7.3 Marking, permanent identification. The inside quarter of each boot shall be marked on the grain side with the correct size and width, the contractor identification symbol, and the month and year (expressed numerically) of the date of contract. This marking shall be impressed into the grain side of the leather in such a manner as to be permanently visible without cutting through at any point, and shall be placed approximately in the center of the inside quarter, and between the two rows of top facing stitching, except that sizes 3 and 3-1/2 on all widths the grain side of the tongue instead of the quarter may be marked not less than 3/4 inch (19 mm) and not more than 1 inch (25 mm) from the top center of the tongue. Figures shall be Arabic and letters Gothic. The figures and letters shall be a minimum of 9/32 inch (7 mm) and a maximum of 3/8 inch (10 mm) in height. The contractor's symbol shall be in a block as shown by the following example: 10 W /AB/ 6-76. he outersole shall be marked as shown on Drawing 2-1-884.

3.7.4 Upper leather fitting. Line marking patterns shall be used for all upper fitting. No die stab marking will be allowed. Quarters shall be closed at the back using stitch type 401 of FED-STD-751. The seam shall extend approximately 1/16 inch (2 mm) on the grain side, and shall be carefully and thoroughly rubbed down. The backseams shall be reinforced on the grain side with backseam tape using stitch type 301. All other upper stitching shall be done using stitch type 301. All upper stitching shall be 8 to 10 stitches per inch (3 to 4 stitches/cm).

3.7.4.1 One piece counterpocket-backstay fitting. The one piece counterpocket-backstay shall be stitched to quarters with two rows of stitching spaced 1/16 to 1/8 inch (2 to 3 mm) between rows on the side edges of the one piece counterpocket-backstay, to the top of the boot or across the top edge of the one piece counterpocket-backstay. The counterpocket-backstay shall be reinforced with two additional inside rows of stitching around the counterpocket and extending to the backstay in conformance with the curvature of the backstay area as indicated by the marker patterns. The inside rows of stitching shall be spaced 1/16 to 1/8 inch (2 to 3 mm) between rows. The use of single or double needle machine is permitted. If inverted "V" seam counterpocket is used, the "V" seam shall be closed, sewing upward from the lasting edge. The closing shall be done with one row of stitching using stitch type 401 and stitched 8 to 10 stitches per inch (3 to 4 stitches/cm). This seam shall extend approximately 1/16 inch (2 mm) on the flesh side and shall be rubbed down. The butt side of this seam shall be on the flesh side of the leather.

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3.7.4.2 Tongue fitting. The quarter eyelet stay shall underlap the tongue and shall be lockstitched with two rows close to the edge of the tongue. The tongue shall be stitched to the vamp grain side out, with two rows of stitching, with one row of stitching not more than 3/32 inch (2 mm) from the edge of the tongue and a second row placed not more than 3/32 inch (2 mm) from the edge of the vamp, both of these rows to include stitching through the vamp lining. The tongue and eyelet facings shall be stitched to the quarters with edge and eyelet rows trimmed flush to 1/8 inch (3 mm) undertrim. The inside top facing, inserted flesh out, shall be fitted overlapping or under the eyelet stay, and shall be undertrimmed and stitched with one row of stitching at the top edge. The bottom edge of the facing shall also be stitched to the quarters with one row of stitching.

3.7.4.3 Eyeletting. On each quarter there shall be nine eyelets for all sizes. On each quarter for all sizes there shall be four eyelets spaced from blucher nose to the bottom portion of the "V" notch of the quarter and the remaining eyelets spaced evenly from top portion of the "V" notch quarter to top of the quarter. The edges of eyelets shall be 1/8 to 5/16 inch (3 to 8 mm) from the edge of the quarter. The eyelets shall be smoothly and securely clinched.

3.7.4.3.1 Lacing for lasting. The machine lacing for lasting shall provide for an opening after lasting of 1 $\pm$ 1/8 inch (25 $\pm$ 3 mm). The quarter shall be laced in the three lower pairs of eyelets. As an alternate, the quarters may be laced in the first and third or the two lower pairs of eyelets.

3.7.4.4 Vamping, barring. Vamping shall be done with two rows of stitching close on the edge of the quarter, plus two rows single or double machine stitching, spaced 5/32 to 1/4 inch (4 to 6 mm) between inside rows. The vamp lap shall extend beyond the second two space rows but not more than 3/16 inch (5 mm). The bar row shall be uniformly spaced between, and parallel to, the two rows of vamping. The vamp bar shall consist of one or two rows and shall be 5/8 $\pm$ 1/16 inch (16 $\pm$ 2 mm) length on each blucher ear. As an alternate, the bar consisting of one row of stitching may be placed outside and above vamping, not more than 1/16 inch (2 mm) and parallel to the vamping row at each blucher ear and in the length specified above. The barring shall be done by single or double needle, or by the automatic method with tying or locking ends.

3.7.5 Insoles. The insoles shall be sorted and cased for even weight. Insoles shall be fleshed and buffed on the flesh side. The insoles shall be rounded to the pattern. Die-cut insoles will be accepted provided they conform to paper insole patterns loaned by the Government. A tolerance of + 0, - 1/16 inch (+ 0, - 2 mm) from Government loaned insole patterns will be allowed around the periphery of the insole.

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3.7.6 Lasting. Uppers may be conditioned by any suitable means except dipping in water. Insoles of the correct size and width shall be tacked to the last with four tacks. One of the four tacks shall be in the center of heel seat, one at shank, one in the ball, and one at the toe. Staples in lieu of tacks may be used for attaching the insole to the last. Care shall be taken that the edge of the insole is flush with or not more than 1/16 inch (2 mm) from the last bottom edge at all points and does not extend beyond the last at any point. The counters shall be assembled in accordance with size schedule so that they will fill the counterpockets. The counter shall be seated to the insole and anchored with a tack near the back end of the counter flange. The tack shall be driven through the counterpocket and insole and clinched against the heel seat plate. Steel tacks may be used for lasting operations. Counters shall be well cemented on both sides. When leather counters are used, the flesh side of the leather counter shall be towards the inside of the boot. The counter shall be turned over the last at the bottom 1/2 ± 1/16 inch (13 ± 2 mm). The correct size of pyroxylin or polystyrene type box toe specified in 3.3.2.2 shall be conditioned and inserted between the vamp lining and the vamp. A light coating of any adhesive specified in 3.3.6.2 shall be applied between the vamp and vamp lining rearward of the box toe. Uppers of the correct size and width shall be assembled to the lasts in a manner to provide for a wiped-in-heel seat of not less than 7/16 inch (11 mm) nor more than 3/4 inch (19 mm). Uppers shall be drawn with care over the lasts with proper tension on the pulling-over machine to assure that the quarters at the blucher points and the vamps are down to the last, and that the blucher noses are even. Care shall be taken that the vamp lining is pulled smooth and tight without any tears. The heel seat shall be wiped-in, tacked and shall be flat and free from wrinkles. The toe shall be firmly and smoothly wiped-in and securely attached by staples. The sides shall be cemented before or in conjunction with side lasting and stapled on a side stapling machine. The staples shall be adjusted and spaced to securely hold the upper in place through the vulcanization operation.

3.7.7 Time allowance on lasts. The boots shall remain on the lasts a sufficient length of time to thoroughly dry the counters and uppers.

3.7.8 Tack and staple pulling. All insole tacks and staples shall be removed with care taken that no broken tack points remain.

3.7.9 Preparation of bottom. The bottom shall be prepared in accordance with the procedures detailed in 3.7.9.1 and 3.7.9.2.

3.7.9.1 Roughing. The edges of the lasted uppers shall be roughed with a cylindrical steel brush, 6.25 ± 0.25 inches (159 ± 6 mm) in diameter and 1 or 2 inches (25 or 51 mm) in width, revolving at a speed of 2500 ± 400 revolutions per minute. The bristles shall have a diameter of 0.015 ± 0.003 inches (0.4 ± 0.1 mm). As an alternate, the bottom roughing may be performed

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with an automatic roughing machine or other suitable automatic roughing equipment. The first stage shall rough the outer edge (featherline) to the extent that the roughed surface will mold to the nip line of the outersole mold but will not be evident above the nip line. The second stage shall rough the remainder of the overlay in such a way as to remove the grain and leave the fibers rough and loose. Wheel roughing prior to wire brush roughing may be performed. Care shall be taken not to rough through the overlay. All loose particles of dust shall be completely removed from the roughed bottom with the use of a stiff bristle brush or compressed air.

3.7.9.2 Skiving and roughing. As an alternate to the procedure detailed in 3.7.9.1, the lasted grain side overlay portion of the quarters and counterpocket shall be skived in a sufficient width to assure the skived surface will mold to the nip line of the outersole mold but will not be evident above the nip line. The scarf edge thickness shall be 1.5 ± 0.3 mm. The entire skived edges, except the area where the stitches overlay the insole, shall be roughed in accordance with 3.7.9.1. The overlay portion of the vamp area shall be roughed in accordance with 3.7.9.1. NOTE: When this alternate method is used, the skiving of the flesh side of the quarters and counterpocket shall not be permitted.

3.7.10 Bottom adhesion. A liberal amount of adhesive as specified in 3.3.6.2 shall be applied with a hand brush to the roughed area and to the exposed part of the insole and the shank. After allowing the adhesive specified in 3.3.6.2 to dry for a minimum of 15 minutes, a second coat of the same adhesive shall be applied to the same roughed area and allowed to dry for a minimum of 40 minutes.

3.7.11 Relasting. The boots shall be relasted using proper size and width metal lasts on the direct molded sole machine. The proper size steel shank shall be attached either while the boot is on the MIL-5 last or on the metal last, with the forward end of the shank approximately 1-1/8 inches (29 mm) rearward of the ball line.

3.7.12 Use of mold release. Coating of the exposed portions of the metal mold with a mold release compound shall be permitted as required.

* 3.7.13 Bottoming. The biscuits for the heel and outersole sections shall be cut from the different batches of rubber, and the outersole biscuit shall extend from toe to rear of heel in one unbroken piece. A cavity for the heel filler shall be cut or otherwise formed in the outersole biscuit to allow for a margin of rubber between the cavity and the outer heel perimeter of 5/8 inch (16 mm) minimum contacting the lasted upper, not including the heel breast line.

* 3.7.13.1 Heel filler. The heel filler shall be inserted in the cavity of the outersole biscuit (see table III).

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* 3.7.13.2 Heel biscuit. The heel biscuit shall be placed in contact with the outersole biscuit so that during the molding operation the two knit together and the heel biscuit forms the bottom of wearing surface of the heel. The contacting surfaces of the heel and outersole biscuits may, when necessary, be lightly brushed with methyl ethyl ketone.

3.7.14 Outersole and heel molding. The outersole and the heel shall be directly molded to the upper of the boot with the use of molding equipment which is approved for this purpose by the contracting officer. The equipment shall be operated so as to exert a minimum pressure of 250 pounds per square inch (1720 kPa) on the compound. The sole and side molds shall be heated to a constant heat of 149° to 163°C (300° to 325°F) and for a dwell time of 15 minutes shall engage the boot for vulcanization (see 4.4.2). The boots shall be molded with the use of metal lasts which shall conform to the measurements contained in MIL-L-41815.

3.7.15 Heel and edge scouring. After removal from the direct molded sole machine, the flash edges shall be removed when necessary by scouring. The use of a knife for trimming spewed flash is prohibited at the upper leather edge.

3.7.16 Tacks or staples. Tacks, nails, or staples, that have been left protruding through or around the insole and cannot be pulled out, shall be cut close to the surface and shall leave no protruding stumps.

3.7.17 Upper finishing.

3.7.17.1 Preparation. The boots shall be cleaned by removing accumulated dirt such as wax or cement, and the surface of the leather shall be conditioned to receive further application of the finish. All thread ends shall be trimmed.

3.7.17.2 Treeing. All wrinkles shall be removed. No material shall be used that may injure the leather or thread.

3.7.17.3 Final finish. The boots shall be repaired and filled. All raw edges, including vamp line, counterpocket-backstay, top and front edges shall be stained to match the upper leather. Materials and methods shall be as specified in 3.3.6.5.1 to 3.3.6.5.3.

3.7.18 Laces. The boots shall be mated. A lace shall be inserted through the top eyelet of the outside quarter of each boot, and the two laces shall be tied firmly together.

3.7.19 Ventilating insole. One ventilating insole as described in 3.3.6.4 shall be inserted in each boot.

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3.7.20 Repairs of stitching.

3.7.20.1 Type 301 and 401 upper fitting stitches. Repairs of type 301 and 401 stitches shall be as follows:

a. When thread breaks or bobbin run-outs occur during stitching, the stitching shall be repaired by restarting the stitching one stitch in back of the end of stitching.

b. Thread breaks, two or more consecutive skipped or runoff stitches noted during inspection of the item (in-process or end item) shall be repaired by overstitching. The stitching shall start one stitch in back of the defective area, and continue one stitch beyond the defective area onto the existing stitching.

* c. Vamp or counterpocket stitching damaged or missing at the junction of the outersole shall be repaired with a minimum of two stitches. Applicable to each row of defective stitching, the first stitch shall be a maximum of 1/4 inch (6 mm) from the junction of the outersole.

NOTE: In lieu of repairing each row of stitching individually, the four rows of stitching in the vamp or counterpocket may be repaired in one operation by stitching across the rows of stitching. Location of stitching shall be as specified above and shall start and end within the rows of stitching. Each group of vamping and counterpocket stitching on a quarter shall be repaired separately.

3.7.21 Replacement of defective components. During the spreading, cutting and manufacturing process, components having material defects or damages that are classified as defects in 4.4.2.2.1, 4.4.2.2.2 and 4.4.3.1.1 shall be removed from production and replaced with non-defective and properly matched components.

3.8 Workmanship. The finished boots shall conform to the quality of product established by this specification. The occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When required (see 6.2), the first article submitted in accordance with 3.2.2, shall be visually inspected as specified in 4.4.3 for compliance with design, construction, workmanship and dimensional requirements.

4.4 Quality conformance inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

* 4.4.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase documents. In addition, testing shall be performed on components and materials listed in table VII. A rubber compound lot shall consist of the amount of outersole and heel rubber compound that is produced or compounded in one day. A representative composite test sample of the lot of rubber shall be vulcanized by using a 10-minute cure at 310°F (154°C) and the composite sample shall be as follows:

(a) Two pieces, 6 by 6 by 0.070 ± 0.010 inches (150 by 150 by 1.78 ± 0.25 mm).
(b) Two pieces, 6 by 6 by 0.250 ± 0.010 inches (150 by 150 by 6.35 ± 0.25 mm).

When the data in the "Number of determinations per sample unit" and the "Results reported as" columns are not specified in table VII, they shall be reported as required by the referenced test methods. All test reports shall contain the individual values utilized in expressing the final result. The lot shall be unacceptable if one or more sample units or the composite sample fails to meet any requirement specified. The sample size shall be as follows:

Lot size	Sample size
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

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TABLE VII. Component tests

Component and lot size expressed in terms of	Characteristic	Requirement paragraph	Test method	Requirement applicable to: Sample unit	Requirement applicable to: Composite	Number of determinations per sample unit	Results reported as: Pass or fail	Results reported as: Numerically to nearest	Sample unit
Box toes (1 box toe)	Material identification	3.3.2.2	1/	-	-	-	-	-	-
	Impregnation compound identification	3.3.2.2	1/	-	-	-	-	-	-
	Thickness	3.3.2.2	1/	-	-	-	-	-	-
Shank, steel (1 pair)	Material identification	3.3.3.1	1/	-	-	-	-	-	
	Thickness	3.3.3.1	1/	-	-	-	-	-	
	Hardness (Rockwell C Scale)	3.3.3.1	E 18 2/	X	-	1	-	Hardness number	2 shanks
	Height, steel rib	3.3.3.1	Visual, micrometer	X	-	1	X	-	
	Length and width	3.3.3.1	1/	-	-	-	-	-	
	Coating identification	3.3.3.1	1/	-	-	-	-	-	
	Finish	3.3.3.1	1/	-	-	-	-	-	
Tacks and staples	Material identification	3.3.3.2	1/	-	-	-	-	-	

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TABLE VII. Component tests (cont'd)

Component and lot size expressed in terms of	Characteristic	Requirement paragraph	Test method	Requirement applicable to: Sample unit	Requirement applicable to: Composite	Number of determinations per sample unit	Results reported as: Pass or fail	Results reported as: Numerically to nearest	Results reported as: Sample unit
Eyelets 1 pound (0.45 kg)	Material identification	3.3.3.3	1/	-	-	-	-	-	
	Thickness	3.3.3.3	1/	-	-	-	-	-	2
	Outside diameter of flange and barrel	3.3.3.3	Gage	X	-	2	-	0.001 inch (0.01 mm)	eyelets
	Diameter of hole	3.3.3.3	Gage	X	-	2	-	0.001 inch (0.01 mm)	
	Overall length	3.3.3.3	Gage	X	-	2	-	0.001 inch (0.01 mm)	
	Finish	3.3.3.3.1	1/	-	-	-	-	-	
Outersole compound (lot)	Abrasive index (before and after aging)	3.3.4.1	D 1630 2/4/	-	X	3	-	Whole number	-
	Hardness (before and after aging)	3.3.4.1	D 2240 2/4/	-	X	5	-	Whole number	-
	Hardness and hardness change at -18°C	3.3.4.1	4.5.1	-	X	1	-	Whole number	
	Cut growth (after aging)	3.3.4.1	D 1052 2/4/	-	X	2	-	One percent	-
	Ozone resistance	3.3.4.1	4.5.2 1/	-	-	-	-	-	-
	Volume swell	3.3.4.1	6001 and 6211 3/	-	X	3	-	One percent	-

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TABLE VII. Component tests (cont'd)

Component and lot size expressed in terms of	Characteristic	Requirement paragraph	Test method	Requirement applicable to: Sample unit	Requirement applicable to: Composite	Number of determinations per sample unit	Results reported as: Pass or fail	Results reported as: Numerically to nearest	Sample unit
Outersole compound (lot) (cont'd)	Polyvinyl chloride exclusion	3.3.4.4	5/	-	-	-	-	-	-
Heel compound (lot)	Abrasive index (before and after aging)	3.3.4.2	D 1630 2/4/	-	X	3	-	Whole number	-
	Hardness (before and after aging)	3.3.4.2	D 2240 2/4/	-	X	5	-	Whole number	-
	Hardness and hardness change at -18°C	3.3.4.2	4.5.1	-	X	1	-	Whole number	
	Ozone resistance	3.3.4.2	4.5.2 1/	-	-	-	-	-	-
	Volume swell	3.3.4.2	6001 and 6211 3/	-	X	3	-	One percent	-
	Polyvinyl chloride exclusion	3.3.4.2	5/	-	-	-	-	-	-
Heel filler (1 filler)	Heel curve radius	3.3.6.1	Gage	X	-	2	X	-	2 fillers of each size in the lot
	Width	3.3.6.1	Gage	X	-	2	X	-	
	Length	3.3.6.1	Gage	X	-	2	X	-	
	Thickness	3.3.6.1	Gage	X	-	2	X	-	
Adhesive	Material identification	3.3.6.2	1/	-	-	-	-	-	-

1/ A certificate of compliance shall be submitted and will be acceptable for the stated requirement.

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2/ Refers to ASTM Standard.

3/ Refers to FED-STD-601.

4/ Aging shall be at a temperature of 100 ± 1°C for 70 hours in accordance with the ASTM D 573 procedure for accelerated aging.

5/ The contractor shall submit a certificate of compliance indicating that the outersole and heel do not contain polyvinyl chloride (PVC).

4.4.1.1 Examination of components.

4.4.1.1.1 Examination of insoles. The insoles shall be examined for visual and dimensional requirements conforming to the provisions of the end item examination of KK-I-570.

• 4.4.1.1.2 Examination of counters. The counters shall be examined for visual and dimensional requirements conforming to the provisions of the end item examination of MIL-C-41814.

4.4.2 In-process inspection. Inspection shall be made at any point or during any phase of manufacturing to determine whether operations or assemblies are accomplished as specified. The contractor shall furnish a certificate of compliance for the outersole and heel molding as specified in 3.7.14 for pressure, heat, and dwell time. The Government reserves the right to exclude from consideration for acceptance any material or service for which in-process inspection has indicated nonconformance.

4.4.2.1 Mold approval. During the molding operation and prior to the formation of the initial inspection lot, one boot shall be randomly selected from the production of each new mold and determination shall be made whether the outersole conforms to the measurements specified in Drawing 2-1-884. If the sole is found to be in conformance, the mold from which it was made shall be approved. If the measurements are found to be nonconforming, an additional five boot outersoles produced from that mold shall be measured for conformance to Drawing 2-1-884 and if all five outersoles conform to the measurements, the mold shall be approved. If all molds used in the processing of the initial inspection lot have been approved, testing of the finished boots for conformance to outersole measurements shall not be applicable. When new molds are added or when molds are repaired, mold approval must be obtained as described previously.

4.4.2.1.1 Identification marking of approved molds. Each mold shall have an identifying number inscribed approximately in the center of the bevelled portion of the heel mold. The mold identification numbering shall be consecutive, beginning with number "1". The height of the numbers shall be not more than 1/8 inch (3 mm).

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4.4.2.2 In-process examination. The defects found during in-process examination shall be classified in accordance with 4.4.2.2.1 and 4.4.2.2.2. The inspection level shall be level II and the acceptable quality level (AQL) shall be 2.5 major and 6.5 total, expressed in defects per one hundred units.

NOTE: Defects designated by an asterisk shall be scored as major when seriously affecting serviceability, and minor when affecting serviceability but not seriously.

4.4.2.2.1 Examination of uppers after all fitting. The upper assembly shall be examined for defects listed below which cannot be seen in the end item. The sample unit shall be one completely fabricated upper assembly prepared for lasting. The lot size shall be expressed in units of one completely fabricated upper assembly for lasting.

Examine	Defect	Classification Major		Minor
Construction and workmanship (general)	Any component missing or not specified type.	X		
	Any component misplaced or not affixed as specified.		*	
	Vamp stitched more than 5/32 inch (4 mm) from edge at throat.	X		
	Vamp stitched more than 3/32 inch (2 mm) from edge at throat but not over 5/32 inch (4 mm).			X
	Vamp throat or wings not skived.	X		
	Counterpocket-backstay or quarter not skived as specified.			X
	Backseam not rubbed down.			X
Quality of leather	Thickness more than 1/2 ounce (0.2 mm) less than minimum specified.	X		
	Thickness less than specified minimum but not exceeding 1/2 ounce (0.2 mm) less.			X
	Thickness more than maximum specified.			X
	Rough fiber on flesh side.		*	
	Off-stretch cut.	X		
	Slaughter cut.		*	

4.4.2.2.2 Examination of boot before last pulling. The partially fabricated boot shall be examined for defects listed below which cannot be seen in the end item. The sample unit shall be one partially constructed boot assembled to the point just prior to pulling off the MIL-5 last. The lot size shall be expressed in units of one partially fabricated boot.

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Examine	Defect	Classification Major		Minor
Bottom of boot	Any component missing or not specified type.	X		
	Upper damaged.		*	
	Insole not scoured on the flesh side as required.	X		
	Poor heel seat, side, or toe lasting.		*	
Upper part of boot	Uppers not firmly pulled down to last.		*	
	Lace openings less than 7/8 inch (22 mm) or more than 1-1/8 inches (29 mm).	X		
	Quarter not laced as specified.			X
	Less than 1/2 inch (13 mm) of upper lasted over insole, except heel seat upper less than 7/16 inch (11 mm) or more than 3/4 inch (19 mm).	X		
	Upper in lasted area not sufficiently roughed or cleaned after roughing.	X		
	Excessive roughing or scouring.		*	

4.4.3 End item inspection.

4.4.3.1 End item examination. The defects found during examination shall be classified in accordance with 4.4.3.1.1. The inspection level shall be II and the AQL shall be 2.5 major and 6.5 total defects per 100 units.

4.4.3.1.1 Visual examination. The boots shall be examined for defects in pairing, design, material, construction, workmanship, finish and marking. The sample unit shall be one completely fabricated boot and the selection shall be by pairs. The inspector shall check to see that ventilating insoles are inserted in the boots. For the pairing examination, and when determining possible difference in outersole thickness, the pair shall be examined together. Each defect found during the examination for pairing shall be scored as a single defect. The lot size shall be expressed in units of one completely fabricated boot. The vamp, and inside and outside quarters of each boot shall be examined for break of leather in accordance with the procedure outlined below:

* (a) To examine the vamp, the boot shall be held in an upright position with both hands. The toe of the boot shall face away from the examiner. Position thumbs on top of vamp approximately halfway between box toe line and blucher noses and 1 inch to 1-1/2 inches (25 to 38 mm) apart. Press downward with thumbs so as to form grain surface into a concave surface and observe the break pattern. Any vamp exhibiting a break pattern at any location comparable to a break pattern represented by numbers 5 through 8 on the Satra scale (see 6.4) shall be scored as a defect. The defect shall be scored regardless of the direction in which the break pattern appears.

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(b) To examine the quarters, the index finger and the second finger held in a "V" position shall be placed against the inside of the quarter, between the fourth and sixth eyelets from top of quarter. The thumb shall be placed on the outside of the quarter over the fifth eyelet. Pressure shall be applied to the quarter with the thumb and fingers so as to impart a break pattern to the leather. The break pattern of the quarter shall be observed and compared with the break patterns of the Satra scale. Any quarter exhibiting a break pattern comparable to a break pattern represented by numbers 5 through 8 on the Satra scale shall be scored as a defect. The following note regarding defects visible at a distance of 3 feet (1 m) is not applicable to examination of the break pattern.

NOTES: Appearance defects shall be scored only when the condition is plainly visible at a distance of 3 feet (1 m) or more.

Defects designated by an asterisk shall be scored as major when seriously affecting serviceability or appearance, and minor when affecting serviceability or appearance but not seriously.

Examine	Defect	Classification Major	Minor
Pairing	Not properly mated, i.e., not right and left of same size.	X	
	Variation in color, luster, or appearance.	*	
	Variation of more than 1/4 inch (6 mm) in height of pair.	X	
	Box toe crooked, long or short.	*	
	Difference in outersole thickness between left and right boots:		
	- more than 3/32 inch (3 mm).	X	
	- more than 1/16 inch (2 mm) but not more than 3/32 inch (3 mm).		X
Cleanness	Any non-removable spot, stain or foreign matter affecting appearance.		X
Color and finish	Not specified color.	X	
	Finish streaky, chipped, or flaky on upper.		X
	Any raw edges not stained to match upper leather.		X
Design, type and style	Not as specified.	X	

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Examine	Defect	Classification Major	Minor
Upper leather	Leather deeply snuffed, i.e., fiber structure damaged.	X	
	Any grub or tick damage, scratches, brands or boney leather.		*
	Fat wrinkles, or veins.		*
	Stretchy vamp.	X	
	Leather embossed or printed.	X	
Break pattern of vamps and quarters	Break pattern comparable to break pattern represented by numbers 5 through 8 on Satra break scale.	X	
NOTE:	Leather exhibiting a break pattern comparable to the numbers indicated above is considered to be loose, pipey, or flanky. The break pattern shall be determined in accordance with 4.4.3.1.1.		
Construction and workmanship (general)	Any cut, tear, hole, repair or factory damage.		*
	Wrinkled or bunched area at the back seam.		*
	Any component or assembly omitted or misplaced, operation omitted or not properly performed (unless otherwise classified herein).		*
	Loose leather fibers along cut edge of quarter vamp line or counterpocket-backstay.		X
Seams and stitching (upper)	Open seam not repaired.	X	
	Repair of open seam not as specified.		X
NOTE:	A seam shall be classified as open when one or more stitches joining a seam are broken or when two or more consecutive skipped or run-off stitches occur. On multiple stitched seams, a seam is considered open when either one or both sides of the seam are open. When the above conditions occur on the inside of the boot it shall be scored as an asterisk (*) defect.		

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Examine	Defect	Classification Major	Minor
Seams and stitching (upper) (cont'd)			
	Loose tension resulting in a loosely secured seam.	*	
	Tight tension resulting in puckering or cutting of leather.	*	
	Wrong stitch or seam type.	X	
	Any row of stitching with less than 8 stitches per inch but not less than 6 stitches per inch		X
	Any row of stitching with less than 6 stitches per inch.	X	
	More than the specified maximum number of stitches.	*	
NOTE:	A plus tolerance of 3 stitches per inch will be allowed when stitching over heavy places or turning sharp corners.		
	Gage of stitching not as specified or irregular.	*	
	Stitching omitted where required.	*	
	Thread ends not trimmed throughout boot.		X
	Needle holes or needle chew.	*	
	Sand hole, i.e., tongue not properly caught in at throat.		X
Vamp or counter-pocket stitching at outersole junction	One or more rows of defective stitching not repaired.	X	
	Repaired but first stitch more than 1/4 inch (6 mm) from outersole junction.		X
	Repaired with one stitch in lieu of two or more stitches.		X
Outersole including heel	Thickness not as specified.	X	
	Flash edges not removed by scouring affecting appearance.		X
	Edge scouring is irregular affecting appearance.		X
	Blister:		
	- 1/2 inch (13 mm) or more in longest dimension.	X	
	- penetrating 1/3 or more the thickness of the rubber.	X	

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Examine	Defect	Classification Major	Minor
Outersole including heel (cont'd)	- more than 3 blisters, more than 1/8 inch (3 mm) but less than 1/2 inch (13 mm) in longest dimension.	X	
	- one, two, or three blisters, more than 1/8 inch (3 mm) but less than 1/2 inch (13 mm) in longest dimension.		X
	Any other defect in tread or edge area.	*	
	Incomplete or incorrect vulcanization of outersole to uppers, i.e., rubber not securely attached.	*	
	Poor knitting between heel and outersole compounds.	X	
Counters	Rolled or curled counter.	*	
	Soft counter.		X
Heel finishing	Heel not finished square and smooth affecting appearance.		X
Eyelets, quarter	Number of eyelets not as specified but each row has the same number.		X
	Not the same number of eyelets in each row.	X	
	Eyelets not properly spaced within the row or misaligned between the rows to an extent interfering with proper lacing.	*	
	Edge of any eyelet less than 1/16 inch (2 mm) from the edge of the quarter.	X	
	Edge of any eyelet less than 1/8 inch (3 mm) but not less than 1/16 inch (2 mm) or more than 5/16 (8 mm) from the edge of the quarter.		X
	Any eyelet not securely clinched.		X
Lining	Vamp lining having excessive fullness, torn, loose, or wrinkled.	*	
Insole	Short or long.	*	
	Any protruding lasting staple or prong.	X	
	Any protruding shank prong on insole.	X	
	Any protruding point of lasting staple or point of tack forward of heel breast line. 1/		
Ventilating insole	Missing, wrong size or wrong side up.	X	

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Examine	Defect	Classification Major	Minor
Marking	Missing, incomplete, incorrect, not applied in the specified manner, misplaced, illegible or not specified size.		X

1/ Any protruding point of lasting staple or point of tack forward of the heel breast line, found in the sample, shall cause rejection of the lot represented.

4.4.3.2 End item testing. The finished boot shall be tested for the characteristics listed in table VIII. The sample unit for all tests shall be one boot. All test results shall contain the individual values utilized in expressing the final result. The lot shall be unacceptable if one or more sample units fail to meet any requirement specified. The sample size shall be five boots regardless of lot size.

TABLE VIII. End item tests of outersole and heel assembly

Characteristic	Requirement paragraph	Test method	Number of determinations per sample unit	Results reported as Pass or fail	Numerically to nearest
Non-marking (outersole and heel)	3.3.4.1 and 3.3.4.2	4.5.3	1	X	-
Bond strength (outersole)	3.3.4.1	4.5.4	1	-	1 pound (10 N)
Measurements 1/	Drawing 2-1-884	Gage	1	X	-

1/ Mold approval (see 4.4.2.1).

4.4.4 Packaging inspection. An examination shall be made to determine that the preservation, packing, and marking comply with the section 5 requirements. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully packaged with the exception that it need not be closed. Examination for the closure defects listed below shall be made on shipping containers fully packaged. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the AQL shall be 2.5 defects per hundred units.

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Examine	Defect
Marking (exterior)	Omitted; incorrect; illegible; of improper size, location, sequence or method of application.
Materials	Any component missing, damaged, or not as specified.
Workmanship	Inadequate application of components, such as: incomplete closure of container flap, loose strapping, inadequate stapling, or improper taping. Bulged or distorted container.
Content	Number of pairs of boots per container is more or less than required.

4.5 Test methods.

* 4.5.1 Hardness and hardness change at -18°C. A specimen 0.250 ± 0.010 inch (6.35 ± 0.25 mm) thick and at least 1 inch (25 mm) wide by 2 inches (51 mm) long shall be tested for hardness as specified in ASTM D 2240 except the scale shall be read 5 seconds after the presser foot is in firm contact with the specimen. The same specimen and the durometer shall then be conditioned for 1 hour at -18 ± 1°C and the hardness then determined at that temperature. The scale shall be read 5 seconds after the presser foot is in firm contact with the specimen. The reading shall be taken as the hardness at -18°C, and the difference between the two determinations shall be recorded as the hardness change.

* 4.5.2 Test for ozone resistance. There shall be no sign of cracking of the compound when tested as follows. Test specimens 1 inch (25 mm) wide, 3-3/4 inches (95 mm) long and 0.070 ± 0.010 inch (1.78 ± 0.25 mm) thick shall be mounted in accordance with the requirements of ASTM D 518 procedure B except that the length of the clamping strips shall be such as to facilitate placement within the test chamber of the ozone cabinet. Test specimens may be taken directly from the outersole. Test specimens for the heel should be molded from the same compound used in fabrication of the heel and cured at the same time and temperature used in manufacture. Samples shall be exposed for 7 days to an ozone concentration of 50 (± 10) parts per hundred million parts of air at a temperature of 40 ± 1° C as described in ASTM D 1149. At the end of the exposure time, the specimens shall be examined under a seven power magnifier.

4.5.3 Test for nonmarking outersoles (including heels). The edge of the outersole and the edge of the heel shall be drawn across a sheet of white bond paper using moderate pressure. If only light marks are made which are easily erased by rubbing with fingers, the outersole and heel shall be classed as nonmarking.

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4.5.4 Bond strength of outersole and upper.

4.5 4.1 Specimen. The test specimen shall be a single finished boot of any size prepared as follows: The boot shall be legibly marked on each side at two points, i.e., at 2 inches (51 mm) and 4 inches (102 mm) from the tip of the toe. The molded outersole shall then be separated from the upper to the 2 inch (51 mm) mark. The separation may be aided by any suitable device or machine capable of withstanding the amount of force necessary to accomplish separation. The use of pincers conforming to type II, class 1 of GGG-N-350, or equivalent, with the sharp cutting edge ground down to a 1/8 inch (3 mm) face, and a last, one or two sizes smaller than the boot, inserted into the boot and mounted on a last stand, have been found to facilitate the preparation of specimens for the bond strength test. After separation to the 2 inch (51 mm) mark is obtained, the boot last, when used, shall be removed and the toe cap of the boot shall be crushed to facilitate mounting of the specimen in the tester. In the event that separation at the 2 inch (51 mm) mark is not between the upper and rubber outersole, that portion of the rubber outersole remaining adhered to the upper shall be separated manually with the aid of a knife or other instrument so as to show separation between leather and rubber outersole. Care should be exercised so that the leather is not cut.

4.5.4.2 Procedure. The apparatus for measuring outersole-upper separation shall be as described in Method 2031 of FED-STD-311. The separated and crushed toe portion of the boot shall be gripped by one clamp in a manner such that the minimum effective jaw surface area applied shall be 1 by 1 inch (25 by 25 mm). (NOTE: With some testing machines modification of the clamps may be required to provide sufficient opening for insertion of the specimen.) The jaw surface shall be centered approximately 1/2 to 1 inch (13 to 25 mm) from the tip of the toe. The other jaw surface shall hold the overlay portion of the separated upper and shall be not less than 1 by 2 inches (25 by 51 mm) with the long dimension perpendicular to the pull of the machine. The separated rubber outersole portion of the boot shall be gripped by the other clamp of the testing machine. Each jaw of the clamp shall be not less than 1 by 2 inches (25 by 51 mm) with the long dimension perpendicular to the pull of the machine. The machine shall be set in operation and the separation continued at a speed of 10 $\pm$ 2 inches (254 $\pm$ 51 mm) per minute until the outersole is separated from the upper to a point past the mark 4 inches (102 mm) from the tip. The maximum value attained during separation shall be taken as the bond strength. When a test specimen separates so that the leather from the upper or the rubber from the outersole tears, that specimen result shall be discarded and a new specimen selected from the lot. For the purpose of Government verification testing, new samples shall not be selected and results shall be based on the remaining specimens.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial as specified (see 6.2).

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5.1.1 Level A. Each pair of properly mated boots shall be tied together by the ends of laces which have been inserted through the top eyelet of the outside quarter of each boot as specified in 3.7.18.

5.1.2 Commercial. Boots shall be unit packed to afford protection against deterioration and physical damage during shipment from the contractor to the first receiving activity. The unit pack and the quantity per unit pack shall be the same as that normally used by the contractor for retail distribution.

5.2 Packing. Packing shall be level A, B or Commercial as specified (see 6.2).

5.2.1 Level A packing. Twelve pairs of boots of one size and width only, preserved as specified in 5.1, shall be arranged as shown in table IX (B) and packed in a fiberboard shipping container conforming to style FOL-L, grade V2s of PPP-B-636. The inside of each container shall be fitted with a box liner conforming to type CF, class weather-resistant, variety DW, grade V15c of PPP-B-636. The approximate inside dimensions of the shipping container shall be as specified in table IX (A). Each container shall have the contents completely covered on the top and bottom with a sheet of 30 pound minimum basis weight (49 g/m^2) kraft paper conforming to type I, grade B of UU-P-268. Each container shall be closed in accordance with method III, water-proofed in accor dance with method V, and reinforced as specified in the appendix of PPP-B-636 except that the inspection shall be in accordance with 4.4.4. Toward the end of the contract or when there are less than the required amounts per container of the same size and width, mixed sizes and widths may be packed in the same container.

TABLE IX. Shipping container dimensions and packing arrangement

(A) Approximate inside dimensions of shipping containers 1/

Range of boot sizes	Length, inches (mm)	Width, inches (mm)	Depth, inches (mm)
5XN through 8-1/2 XW	30-1/2 (775)	12-1/2 (318)	13 (330)
9XN through 12XW	30-1/2 (775)	13-1/4 (337)	13-3/4 (349)

1/ Approximate dimensions are furnished as a guide only. For boot sizes below 5XN and above 12XW, the contractor shall determine the size case to use. The dimensions of the case shall be such that it will result in a snug fit of the contents to be packed therein. The use of blocking materials to accommodate small size boots in an oversize container will not be permitted.

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(B) Packing arrangements. Arrangement of boots within the shipping container shall be in four layers of three pairs per layer on their sides with the first layer started by placing the left boot in the left-hand end of the container with the heel toward the front and the sole facing left, and the right boot laid over the quarter of the left boot in the same manner. Two more pairs shall be placed in a like manner to complete the first layer. Care shall be taken that the first layer is properly spaced in order to facilitate the packing of the other three layers. The second layer shall start at the left by placing the right boot with the heel toward the rear and the top facing left and the sole face to face with the heel of the second boot in the first layer, and so on to complete the layer. The third layer shall be placed exactly as the first, and the fourth layer shall be the same as the second. The last boot placed in the first and third layers and the first boot placed in the second layer shall have the tops turned upward. The first boot placed in the top layer shall have the top folded under. (See figure 1).

5.2.2 Level B packing. Twelve pairs of boots of one size and width only, preserved as specified in 5.1, shall be arranged as shown in table IX (B) and packed in a fiberboard shipping container conforming to style FOL-L, type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636. The manufacturer's joint shall be metal stitched. The inside of each container shall be fitted with a box liner conforming to type CF, class domestic, variety DW, grade 275 of PPP-B-636. The approximate inside dimensions of the shipping container shall be as specified in table IX (A). Each container shall have the contents completely covered on the top and bottom with a sheet of 30 pound minimum basis weight (49 g/m^2) kraft paper conforming to type I, grade B of UU-P-268. Each container shall be closed in accordance with method II as specified in the appendix of PPP-B-636 except that the inspection shall be in accordance with 4.4.4. Toward the end of the contract or when there are less than the required amount per container of the same size and width, mixed sizes and widths may be packed in the same container.

5.2.2.1 Weather-resistant fiberboard containers. When specified (see 6.2), the shipping container shall be a grade V3c, V3s or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with method III as specified in the appendix of PPP-B-636 except that the inspection shall be in accordance with 4.4.4.

5.2.3 Commercial packing. The boots, preserved as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery to destination at the lowest transportation rate for such supplies. The quantity per shipping container shall be the same as that normally used by the contractor for retail distribution. Containers shall be in accordance with Uniform Freight Classification or National Motor Freight Classification, as applicable.

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5.3 Marking. In addition to any special marking required by the contract, shipping containers shall be marked in accordance with MIL-STD-129.

5.3.1 Width marking. Following the nomenclature and size designation, in the same size lettering, the width shall be spelled out, i.e., EXTRA NARROW, NARROW, REGULAR, WIDE, OR EXTRA WIDE.

5.3.2 Labels, mixed sizes. Each shipping container packed with mixed sizes and widths shall have securely attached to the side and end directly under the printing or stenciling, a white paper label 5 by 4 inches (130 by 100 mm) with the words MIXED SIZES AND WIDTHS clearly stamped or printed thereon and under these words shall be legibly stamped or printed the correct quantity of pairs, sizes, and widths contained therein.

6. NOTES

6.1 Intended use. The boots are intended for use by military personnel of the Department of Defense.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Size and width required (see 1.2).
(c) When laboratory report approval is not required (see 3.2.1).
(d) When a first article is required (see 3.2.2, 4.3 and 6.5).
(e) Selection of applicable levels of preservation and packing (see 5.1 and 5.2).
(f) When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).

6.3 Samples. For access to samples, address the procuring activity issuing the invitation for bids.

6.4 Satra scale. The Satra scale may be obtained from the British Shoe and Allied Trade Research Association, Satra House, Kettering, Northants, England or may be obtained from Satra House's North American agent: Bata Engineering, Batawa, Ontario, Canada.

6.5 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of ASPR 7-104.55. The first article should be a preproduction sample. The first article should consist of one pair of finished boots in any size and width. The contracting officer should include specific instructions in all procurement instruments, regarding arrangements for inspection and approval of the first article.

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* 6.6 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the specification (see 3.3).

6.7 Metric equivalents. Metric equivalents, indicated in parentheses throughout this document, are based on practices, conversion factors, and symbols, specified in ASTM E 380 Standard for Metric Practice, and are for information only. In each instance, the value stated in US customary units shall be controlling.

6.8 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:

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Air Force - 99

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Preparing activity:

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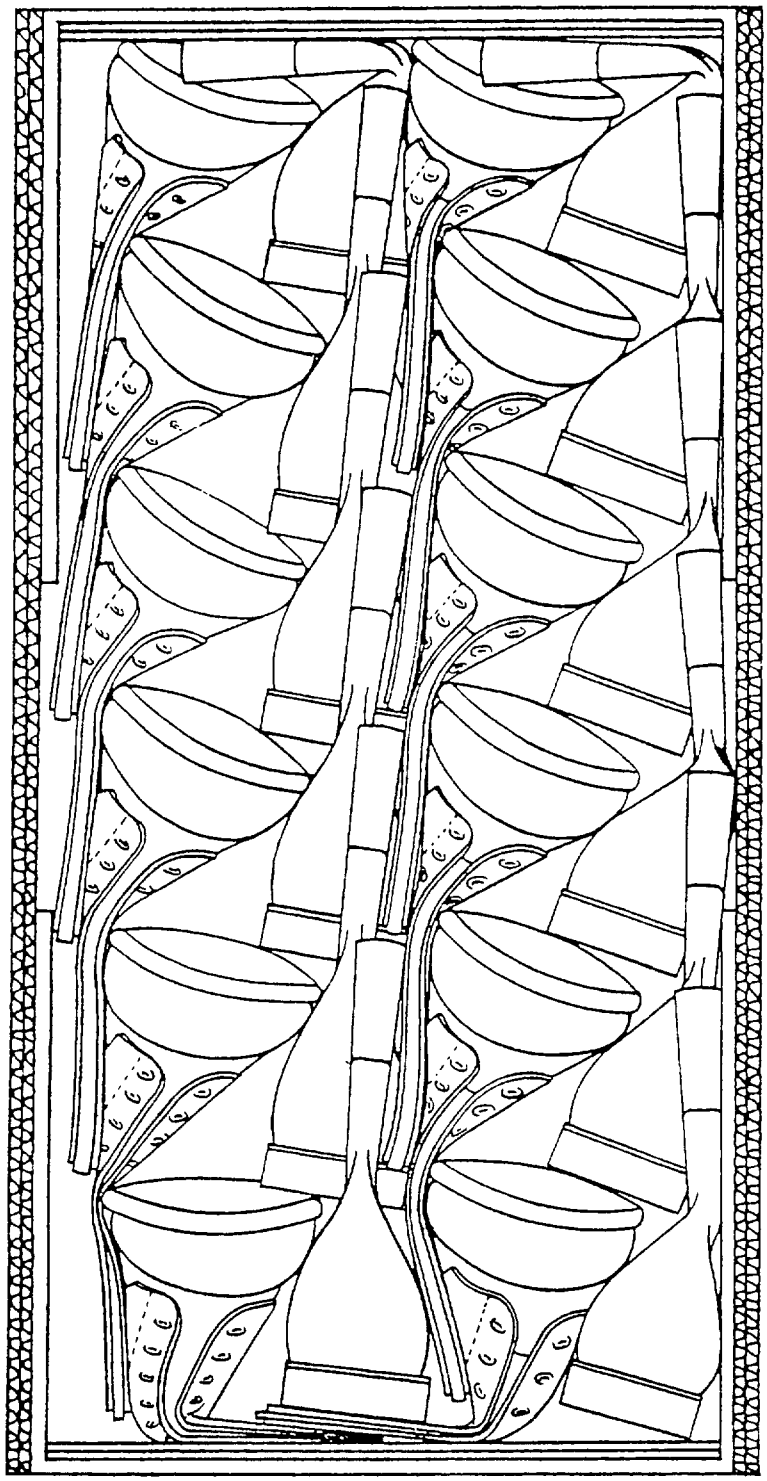


FIG. 1 BOOT ARRANGEMENT WITHIN SHIPPING CONTAINER (SIDE REMOVED)
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5. PROBLEM AREAS

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b. Recommended Wording:

c. Reason/Rationale for Recommendation:

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7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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